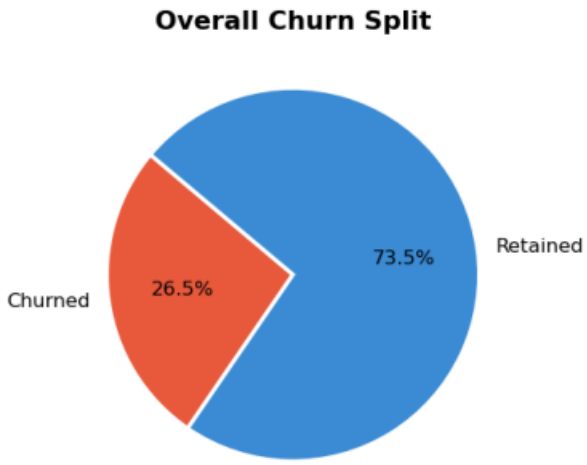
Customer Churn Analysis

Telco Dataset — Exploratory Data Analysis Report

Total customers:	**7,043**
Overall churn rate:	**26.5%**
Highest-risk segment:	**Month-to-month contracts**
Charts included:	**4**

GitHub: customer-churn-analysis / eda-insights



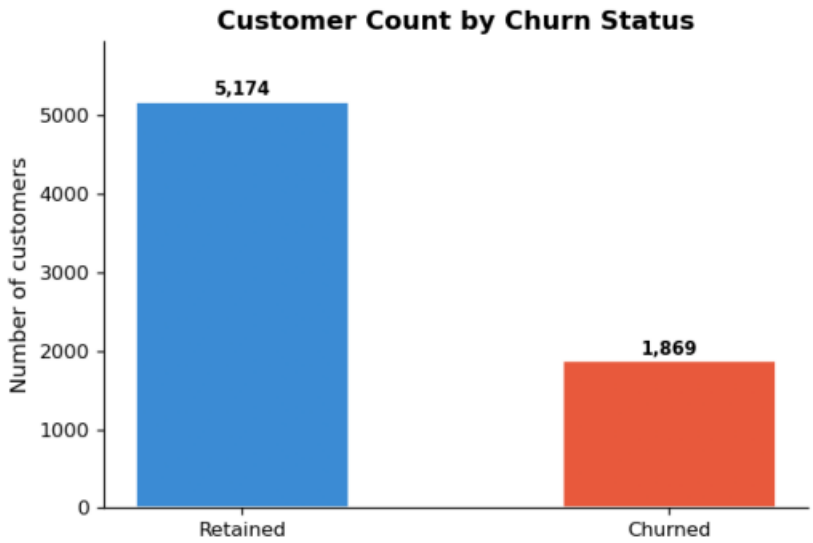


Fig 1 — Overall Churn Split

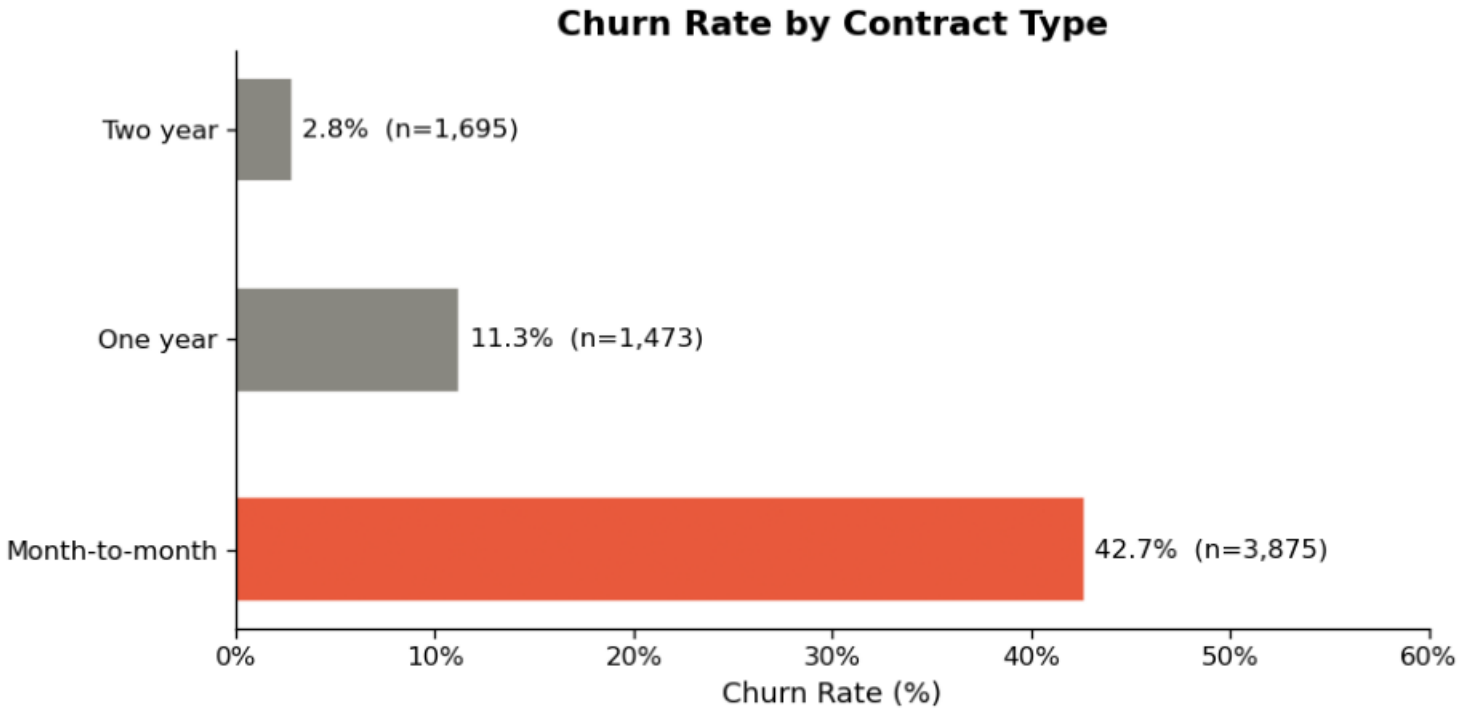


Fig 2 — Churn Rate by Contract Type

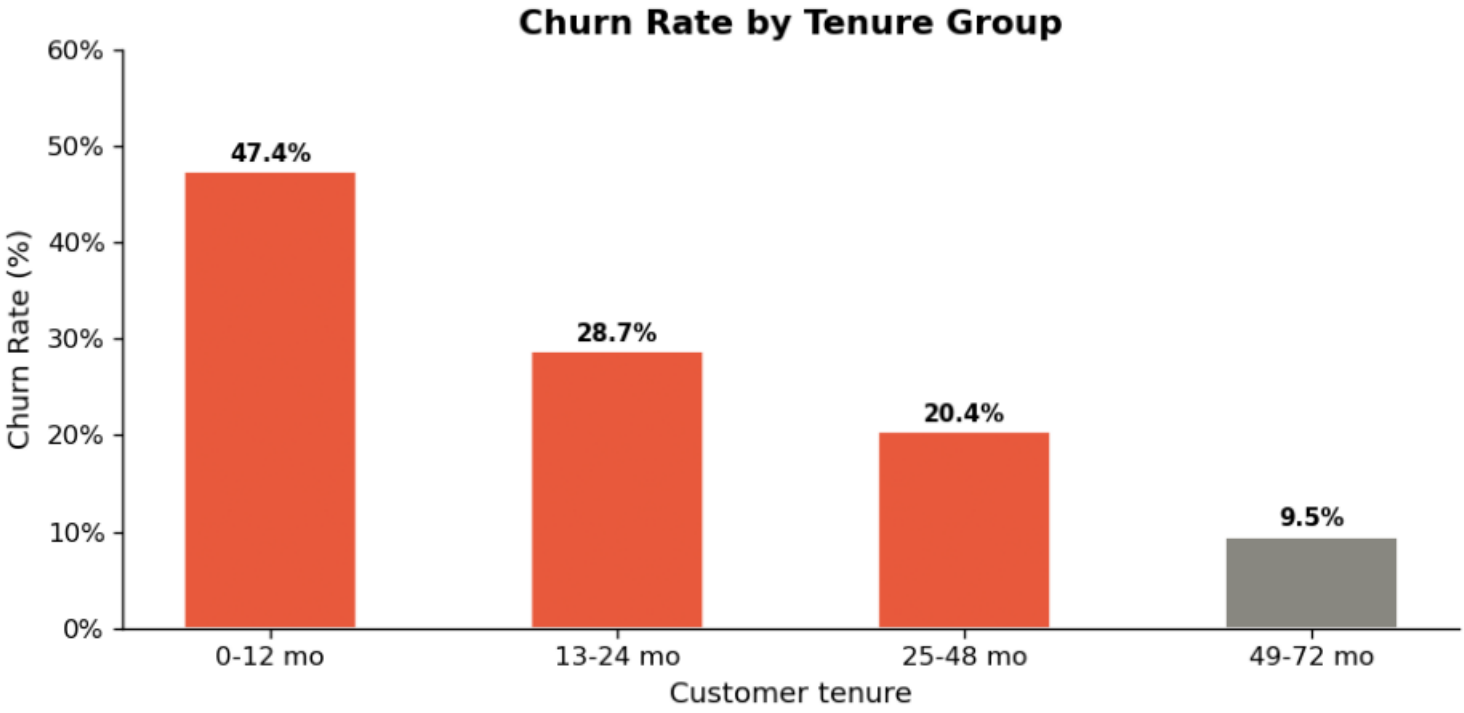


Fig 3 — Churn Rate by Tenure Group

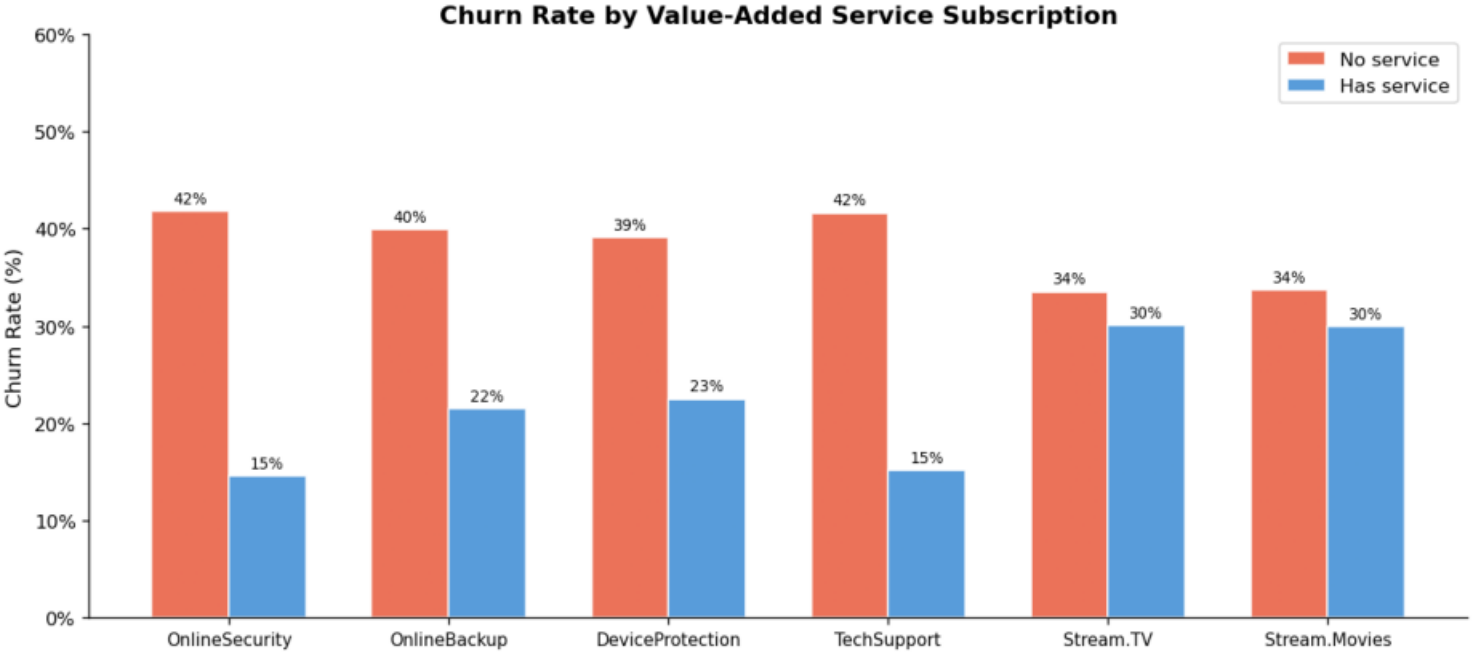


Fig 4 — Monthly Charges Distribution

Key Insights & Recommendations

1. **Contract type:**
   Month-to-month churn 43% vs 2-year churn 3%

2. **Tenure:**
   New customers (0-12 mo) are highest-risk group

3. **Monthly charges:**
   Churned customers pay 74/*moavgvs*61/mo

4. **Internet:**
   Fiber optic 42% vs DSL 19% churn

5. **Payment:**
   Electronic check 45% churn — highest of all methods

6. **Senior citizens:**
   42% churn rate vs 24% non-senior

7. **Online security:**
   Customers without it churn 2× more

8. **Gender:**
   No meaningful impact on churn

Top Recommendations

1. Incentivise annual contracts — month-to-month churn is 43%

2. Launch early-tenure onboarding (0-6 months) — highest ROI window

3. Nudge electronic check payers to auto-pay — 45% vs ~16% churn